# Field Case Study Report of "Avare Biotech" app powered by MAKSA technology on Fresh Dog Semen (Ontario, Canada, 2025)

**Date: December 2025**
**Location: Ontario, Canada**

| | |
|---|---|
| Conducted by:<br><br>Rasa Levstein (Ontario, Canada) | Verified by:<br><br>Avare Biotech Team (Dubai, UAE) |
| **Owner/Theriogenologist at Graham Animal Hospital (Canada)** | Maksim Mershiev,<br>The CEO&Co-Founder of Avare Biotech |
| | |
| | |
| Signature: | Signature: |

## Executive Summary / Conclusions

- Species-specific validation, reproductive specialist involvement, and iterative refinement before market readiness are required for the MAKSA technology to enter canine industry.
- Concentration and motility results for canine semen analysis are inconsistent rather than reliable at this stage, indicating the system is not yet ready for clinical use in this species. **The primary limitation is the system's high sensitivity to input quality, particularly video recordings at 200x microscope magnification and 3x smartphone zoom are vital.** Variations in capture conditions directly affect output consistency.
- This reinforces the necessity of strict adherence to veterinary recording instructions and further refinement of species-specific protocols before broader validation.

## Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for **fresh canine semen** evaluation against conventional CASA

## Materials and Methods

- A standard lab approach to collect and evaluate fresh canine semen was used
- The horse species was selected in the app because its semen size is more comparable to dogs, which improves cross-species adaptation.

*Table 1 MAKSA results comparison*

| Tests done | Key Semen Parameters | MAKSA | | | Nucleo-Counter | Manual analysis | Difference, % |
|---|---|---|---|---|---|---|---|
| Test 1 | Concentration M/ml | 22.4 | 49.6 | 129.6 | 104 | | 105% |
| | Total Motility % | 62 | 26 | 41 | - | 80 | 100% |
| | Progressive Motility % | 35 | 22 | 29 | - | | - |

## Discussion / Expert Insights

- Motility and Concentration values are different and require adjustment via training the software with fresh-diluted canine semen data (videos).
- Videos impact the results severely

## Recommendations

- Train the app on specific canine data set
- Improve input video recordings (guide)

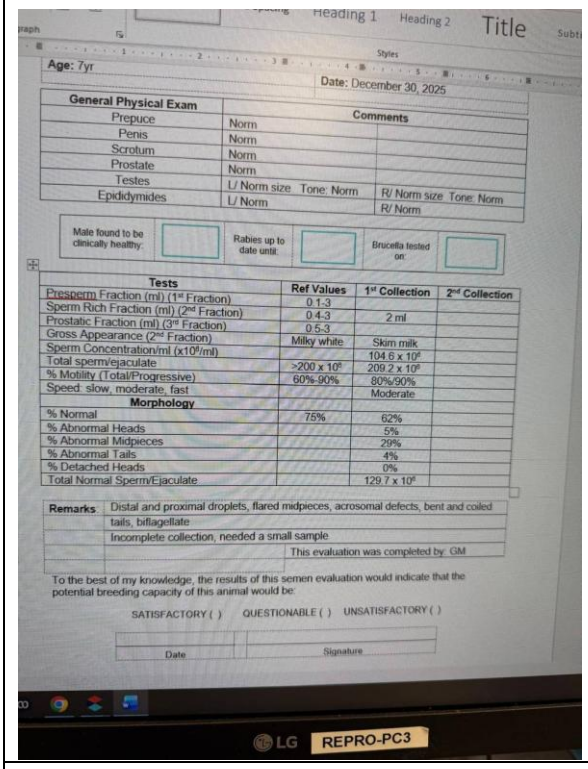
## Appendix

Lab report results and MAKSA PDF report samples

AVAREBIOTECH

Avare Biotech | @Avareit | @avare_biotech | @AvareBiotech

## Dog 1: A final lab report



Age: 7yr

Date: December 30, 2025

| General Physical Exam | Comments | |
|---|---|---|
| Prepuce | Norm | |
| Penis | Norm | |
| Scrotum | Norm | |
| Prostate | Norm | |
| Testes | L/ Norm size Tone: Norm | R/ Norm size Tone: Norm |
| Epididymides | L/ Norm | R/ Norm |

Male found to be clinically healthy: | Rabies up to date until: | Brucella tested on:

| Tests | Ref Values | 1st Collection | 2nd Collection |
|---|---|---|---|
| Presperm Fraction (ml) (1st Fraction) | 0.1-3 | | |
| Sperm Rich Fraction (ml) (2nd Fraction) | 0.4-3 | 2 ml | |
| Prostatic Fraction (ml) (3rd Fraction) | 0.5-3 | | |
| Gross Appearance (2nd Fraction) | Milky white | Skim milk | |
| Sperm Concentration/ml (x10⁶/ml) | | 104.6 x 10⁶ | |
| Total sperm/ejaculate | >200 x 10⁶ | 209.2 x 10⁶ | |
| % Motility (Total/Progressive) | 60%-90% | 80%/90% | |
| Speed: slow, moderate, fast | | Moderate | |
| **Morphology** | | | |
| % Normal | 75% | 62% | |
| % Abnormal Heads | | 5% | |
| % Abnormal Midpieces | | 29% | |
| % Abnormal Tails | | 4% | |
| % Detached Heads | | 0% | |
| Total Normal Sperm/Ejaculate | | 129.7 x 10⁶ | |

Remarks: Distal and proximal droplets, flared midpieces, acrosomal defects, bent and coiled tails, biflagellate

Incomplete collection, needed a small sample

This evaluation was completed by: GM

To the best of my knowledge, the results of this semen evaluation would indicate that the potential breeding capacity of this animal would be:

SATISFACTORY ( ) QUESTIONABLE ( ) UNSATISFACTORY ( )

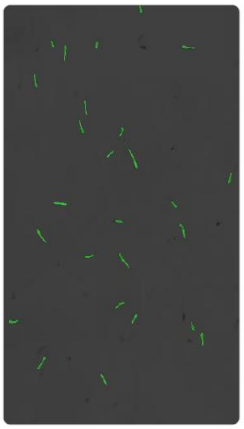
Date | Signature

LG REPRO-PC3

## Dog 1: MAKSA values

**Analysis 30.12.2025**

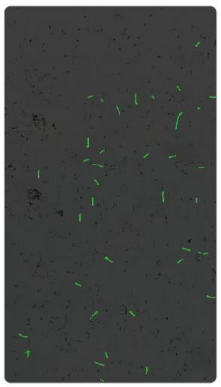
**Animal:** — **Name:** Dog 2 **Age:** 7 **Breed:** Bullmastiff **Status:** Ready **Created:** 2025-12-30 19:24



| | |
|---|---|
| Total objects: | 12 |
| Concentration (millions/ml^3): | 19.20 |
| Dilution rate: | [1:2] |
| Volume of field of view (mm^3): | 0.005060 |
| Progressive Motility: | 35% |
| Non-Progressive Motility: | 27% |
| Immotility: | 38% |

**Analysis 30.12.2025**

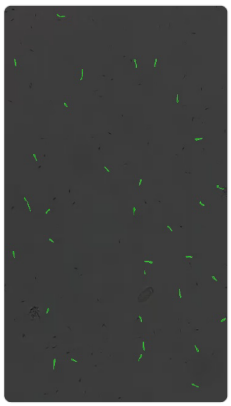
**Animal:** — **Name:** Bauer **Age:** 7 **Breed:** bullmastiff **Status:** Ready **Created:** 2025-12-30 18:42



| | |
|---|---|
| Total objects: | 81 |
| Concentration (millions/ml^3): | 129.60 |
| Dilution rate: | [1:1] |
| Volume of field of view (mm^3): | 0.007262 |
| Progressive Motility: | 29% |
| Non-Progressive Motility: | 12% |
| Immotility: | 59% |

**Analysis 30.12.2025**

**Animal:** — **Name:** Dog **Age:** 7 **Breed:** Bullmastiff **Status:** Ready **Created:** 2025-12-30 19:05



| | |
|---|---|
| Total objects: | 31 |
| Concentration (millions/ml^3): | 49.60 |
| Dilution rate: | [1:2] |
| Volume of field of view (mm^3): | 0.007098 |
| Progressive Motility: | 22% |
| Non-Progressive Motility: | 4% |